

Cognitive Decision Support System Detailed Team Task Assignment Guide

PROJECT TITLE:

Cognitive Decision Support System Using Logistic Regression + FLAN-T5 Small

PROJECT GOAL:

Convert messy human thoughts into:

Decision Prediction → Outcome Simulation → Action Plan → Smart Chatbot Response

Example:

User Input:

I am confused between higher studies and getting a job. Family pressure is high.

System Output:

Take a job first for short-term financial stability while preparing for higher studies later.

This is not a normal chatbot.

This is a Decision Support System.

TEAM MEMBER DIVISION (4 MEMBERS)

MEMBER 1 = ML MODEL + DATASET + LOGISTIC REGRESSION

MEMBER 2 = NLP + PREPROCESSING + EMOTION DETECTION

MEMBER 3 = SIMULATION ENGINE + ACTION PLANNER

MEMBER 4 = UI + GITHUB + DEPLOYMENT + FINAL TESTING

Everyone must complete their own module.

No one should wait for others.

MEMBER 1 DETAILS

(MAIN BRAIN OF THE PROJECT)

ROLE:

Machine Learning + Dataset + Logistic Regression

LANGUAGE REQUIRED:

Python

LIBRARIES REQUIRED:

pandas

numpy

scikit-learn

pickle

TASKS TO COMPLETE:

TASK 1:

Create dataset.csv

Columns:

input_text

target_label

target_output

Example:

input_text:

I am stressed about exams and family pressure

target_label:

Study First

target_output:

Prioritize studies first and reduce distractions to improve academic performance.

Target:

Minimum 500 rows first

Then improve to 1000+

TASK 2:

Clean dataset

Check:

No blank rows

No repeated rows

Correct labels

Good response quality

TASK 3:

Train Logistic Regression Model

Purpose:

Predict decision category

Possible Labels:

Study First

Job First

Higher Studies

Balanced Plan

Counseling Needed

TASK 4:

Save model

Final file:

model.pkl

TASK 5:

Create prediction function

Example:

Input:

Family needs money urgently

Output:

Prediction:

Job First

FINAL DELIVERABLES:

dataset.csv
model.pkl
predict.py

MEMBER 2 DETAILS (NLP UNDERSTANDING LAYER)

ROLE:
Text Processing + Emotion Detection + Sentiment Analysis

LANGUAGE REQUIRED:
Python

LIBRARIES REQUIRED:
spaCy
TextBlob
NLTK
pandas

TASKS TO COMPLETE:

TASK 1:
Text Cleaning

Do:
lowercase conversion
remove punctuation
remove unnecessary spaces

TASK 2:
Tokenization

Break sentences into words

Example:
I feel stressed about exams

becomes

I / feel / stressed / about / exams

TASK 3:
Stopword Removal

Remove words like:
is
the
am
are

TASK 4:
Keyword Extraction

Find important words like:
exam

money
career
family pressure

TASK 5:
Emotion Detection

Possible Emotions:
Stress
Anxiety
Confusion
Depression
Motivation

TASK 6:
Sentiment Analysis

Positive
Negative
Neutral

Example Output:
Emotion = Stress
Sentiment = Negative
Keywords = exams, money

FINAL DELIVERABLES:
preprocess.py
emotion_detection.py

MEMBER 3 DETAILS
(LOGIC + DECISION IMPACT LAYER)

ROLE:
Simulation Engine + Action Planner

LANGUAGE REQUIRED:
Python

LIBRARIES REQUIRED:
Python basic logic
optional pandas

TASKS TO COMPLETE:

TASK 1:
Create Simulation Engine

After prediction:
simulate future outcomes

Example:

Decision = Job First

Short-term:
Financial stability

Long-term:
Higher studies delayed

Risk:
Academic gap

Benefit:
Reduced family pressure

TASK 2:
Create Rule Mapping

Different decision = different outcomes

TASK 3:
Create Action Planner

Generate weekly plan

Example:

Week 1:
Prepare Resume

Week 2:
Apply for internships

Week 3:
Entrance exam planning

Week 4:
Future study roadmap

TASK 4:
Return final practical suggestions

This part makes the project useful.

FINAL DELIVERABLES:
simulation.py
planner.py

MEMBER 4 DETAILS
(FACE OF THE PROJECT)

ROLE:
Streamlit UI + GitHub + Deployment + Final Testing

LANGUAGE REQUIRED:
Python

LIBRARIES REQUIRED:

streamlit
matplotlib
GitHub
Hugging Face Spaces

TASKS TO COMPLETE:

TASK 1:
Create Streamlit Dashboard

Add:
Input text box
Submit button
Emotion display
Prediction display
Simulation results
Action plan display

TASK 2:
Improve UI

Make it clean and professional

Faculty should feel:
this looks serious

TASK 3:
Create GitHub Repository

Upload:
all files
requirements.txt
saved model

TASK 4:
Deploy Online

Best Free Option:
Hugging Face Spaces

Steps:
Create account
Create new Space
Choose Streamlit
Connect project
Deploy

TASK 5:
Final Testing

Check:
all modules work together

FINAL DELIVERABLES:
app.py
requirements.txt

GitHub repo
Hosted project link

IMPORTANT PROJECT ORDER

STEP 1:
Dataset First

STEP 2:
Logistic Regression

STEP 3:
NLP Module

STEP 4:
Simulation Engine

STEP 5:
Action Planner

STEP 6:
Streamlit UI

STEP 7:
Deployment

STEP 8:
FLAN-T5 Upgrade

Never start with UI first.
Never start with FLAN-T5 first.

WHEN TO USE FLAN-T5

ONLY after Logistic Regression project works fully.

Do not replace Logistic Regression.

Use both.

Logistic Regression:
Decision Prediction

FLAN-T5:
Human-like Response Generation

This is the strongest architecture.

STRONG VIVA LINE

Say:

Logistic Regression is used for structured decision classification, while FLAN-T5 Small is used for context-aware natural language response generation.

This answer is very strong.

FINAL RULE

Make it work first.

Make it smart second.

A working regression project is better than a broken transformer project.